

Internalize or Retrieve: Online Memory Placement for Test-Time Continual Learning Agents

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Abstract

Deployed LLM agents adapt mainly through prompting, retrieval, and external tools; their weights stay frozen, so nothing they experience becomes internal knowledge. Parametric knowledge editing offers a path to internalization, but existing work treats every incoming fact as an edit and evaluates it only with question-style probes. We argue the missing component is the placement decision itself: at test time, an agent must decide, per memory candidate, whether to internalize it into model weights, keep it in a reversible retrieval store, or discard it. We present MEMPLACE, a serving-time memory architecture built around this decision: typed routing writes durable beliefs as multi-key codebook edits while reference facts live in a retrievable store with a shared dedup/supersede lifecycle, and a probe–elicit–compose read path makes internalized knowledge usable in open tasks with per-answer attribution. On a multi-session personal-agent workload with hidden type labels and supersession events, evaluated primarily in a real-capacity setting (no entry-level eviction; codebook and retrieval stores in deployment hold tens of thousands of records), a type-aware single-store baseline with replace-and-drop scoring 0.731 overtakes the LLM router (0.708) and all-retrieve (0.624), internalization shows no composite advantage (0.401 under a persistence stress test; all-retrieve recall 0.631 overtakes the router at 0.568 once eviction is removed), and dual-write collapses (0.226). Lifecycle value sits in replace and drop, not in editing per se. A persistence stress test with a budgeted store still has the router matching oracle within run-to-run variance (0.612 vs. 0.639); a counterfactual per-memory utility regression fails to recover oracle placement—pointing to sequence-aware placement learning as the open problem.

1 Introduction

Agents deployed today adapt through three channels that never touch their weights: prompting, retrieval, and tool use. The memory-agent line [Packer et al., 2023, Zhong et al., 2024, Chhikara et al., 2025] externalizes everything—the next answer depends on retrieval hits, prompt budget, and chunk ordering. The knowledge-editing line [Meng et al., 2022, 2023, Hartvigsen et al., 2023, Fang et al., 2026] provides the machinery to write knowledge into weights, but treats every incoming item as an edit and evaluates the result with question-style probes, so internalized knowledge is almost never observed in open generation. Between the two sits a decision neither line makes: *placement*—for each memory candidate arriving at test time, choose {internalize into weights, keep in a reversible store, discard}.

This paper makes placement a first-class object. We instantiate it in MEMPLACE, a serving-time memory architecture whose write path types and routes candidates (durable beliefs → multi-key codebook edits; reference facts → a retrievable store with dedup/supersede lifecycle; transients → discard) and whose read path makes internalized knowledge usable beyond question answering: a planner decomposes an open task into explicit probes, a codebook gate decides which probes the weights should answer, answers are elicited parametrically with retrieval disabled, and composition attributes each answer to its memory.

We evaluate placement with a workload built to make both extremes fail somewhere: 4 synthetic users, 210 typed memory candidates with hidden labels, 740 probes spanning immediate, delayed, paraphrased, open-scenario, supersession, and near-miss forms. Seven arms share one lifecycle and differ only in the placement decision. The main reading is a real-capacity setting with eviction disabled (Table 5); a budgeted store is retained as a persistence stress test (Table 1), not as a deployment claim.

Claim 1 (persistence stress test). On the budgeted-store matrix—oldest-first eviction plus ambient distractors, read as a stress test rather than deployed capacity—the LLM router reaches 0.612 composite versus oracle 0.639, a gap of 0.027 inside the rerun drift bound (0.025). Its lead over the better extreme is 0.102, four times that bound.

All-internalize fails on reference facts and locality; all-retrieve fails on evicted items and stale transients; dual-write is worst at 0.226 (Table 6).

Claim 2 (the lifecycle signal). Lifecycle value lives in two operations—*replace* (same-subject supersession) and *drop* (transients)—not in editing per se. In the real-capacity setting, a type-aware single-store arm that retrieves beliefs and facts, drops transients, and never edits scores 0.731 and is the strongest reported arm; internalization shows no composite advantage (all-internalize 0.401 on the stress-test matrix; without eviction, all-retrieve recall 0.631 overtakes the router’s 0.568). Freshness at oracle level is delivered by store-side record replacement (§4.2). This S8-versus-router comparison is well-posed because both arms run the same replace/drop lifecycle; deployed codebooks and retrieval indexes hold tens of thousands of records, so entry-level eviction at ~ 50 memories per user is not a deployment mechanism. A counterfactual per-memory utility router, trained on isolated edit-vs-retrieve clones, collapses to near-all-retrieve. Dual-write degrades both channels simultaneously. Learning placement from outcomes therefore requires sequence-aware replace/drop signals, which we leave as future work.

2 MEMPLACE

Write path. Each user turn yields typed memory candidates: *beliefs* (durable stances, preferences, habits with no mutable external referent), *facts* (entries with concrete, updatable referents—dates, places, providers, accounts), and *transients* (one-off states and context). An LLM router types each candidate; a fixed type-to-store map routes it. Beliefs are internalized as multi-key codebook edits: alongside the native key, answer-free canonical key prompts are written as additional retrieval aliases for the same trained value, which separates near-colliding personal-belief queries. Facts enter a retrievable store. A single dedup/supersede lifecycle runs for every arm: a newer same-subject memory supersedes the older one in whichever store(s) hold it—all arms in this paper share this lifecycle and differ only in the placement decision. Transients are discarded.

Editing backend. Internalization uses HoReN [Fang et al., 2026], a parameter-preserving codebook editor that wraps one MLP projection with a discrete key–value memory and retrieves by normalized Hopfield similarity; we adopt it unchanged (hyperparameters locked to the published configuration) and treat its published properties—base weights untouched, sequential stability at scale—as confirmatory context for our unrelated-pool results (§4.4).

Read path. Direct questions are answered from the routed store(s). Open tasks go through probe–elicit–compose: an LLM planner turns the task text into 2–4 user-centric probes (mechanically checked to contain no candidate-answer words); each probe passes the codebook gate at a frozen threshold; gated probes are answered *parametrically*—generation against the edited weights with retrieval disabled—while the same probes also query the store; collected notes compose the final output. Parametric elicitation is a design choice for attribution and persistence rather than an empirically demonstrated advantage: at our dev ablation scale ($n=19$) text-injection performs identically on usage. Every arm shares this read path.

Serving. Consolidation can run on a shadow copy of the editor and be promoted atomically at a request boundary; we describe this architecture for completeness—the experiments here are single-model and make no hot-swap claims.

3 Experimental Setup

Workload. 4 synthetic users, with 210 typed candidates in total across session streams with hidden labels; 740 probes of seven kinds (immediate/delayed/paraphrase QA, free scenarios bundling 2–3 memories, supersede-old/new, near-miss twins). Personae and memories are LLM-generated fictional users, disclosed as such; PersonaChat [Zhang et al., 2018], MSC [Xu et al., 2022], and LoCoMo [Maharana et al., 2024] serve as structural and typological references, not data sources. Probes are machine-checked target-free; generator, router, planner, and judge prompts were frozen with hashes before test evaluation.

Arms. All-retrieve (S1), all-internalize (S2), random (S3), oracle-from-labels (S4), LLM router (S5), counterfactual-utility router (S6, preliminary), and dual-write (S7), plus a type-aware single-store defensive arm (S8: beliefs and facts

both retrieved, transients discarded, zero edits) added after the main matrix to answer “why not retrieval-only?”—all disclosed as appended arms where applicable.

Scoring. A frozen per-(type $\times$ probe-kind) semantics matrix maps every probe to recall/freshness/locality axes; transients never enter the recall composite, and delayed transient probes are reverse-scored (asserting a stale state is the failure). The composite is the equal-weight axis mean; probe-level bootstrap CIs and per-user splits accompany the main table. Judged naturalness (a frozen judge prompt, different model family from all system LLMs) is always paired with usage rates.

Retrieval pressure. Two knobs model *read-side noise* and keep their deployment reading: top- k caps retrieved hits, and pinned distractor documents are the environmental non-answer material any real store accumulates. A third knob—oldest-first eviction under a store budget—is a *persistence stress test*: deployed HoReN codebooks and retrieval indexes hold tens of thousands of records, so entry-level eviction at this workload’s scale is not a deployment mechanism. The two families do not share a budget (distractors are never evicted). The main reading disables budget and eviction (pressure-off) with everything else frozen; the budgeted matrix is reported in full and labeled as the stress test.

4 Results

4.1 Real-capacity setting

The main reading disables budget and eviction (Table 5). All-retrieve’s recall recovers to 0.631, overtaking the router’s 0.568: internalization’s recall benefit does not appear once entry-level eviction is removed. The type-aware single-store arm (S8) scores 0.731, above the router at 0.708. Freshness at oracle level is delivered by store-side replacement, not by writing beliefs into weights. We read this as the real-capacity world: deployed stores are not entry-starved.

4.2 Why not retrieval-only?

The natural defensive baseline is type-aware retrieval-only (S8): route by type, drop transients, never edit, keep the shared replace/drop lifecycle. S8 is a *strong default* at this scale: 0.731 without eviction and 0.618 on the persistence stress test (Table 5), at or above the router at both states, with freshness at oracle level. Axis-level slices (Appendix C) show the same pattern on fact supersession; belief $\times$ supersede and belief $\times$ near-miss cells are empty ($n=0$) because those pairs in the frozen workload fall on facts. Residual deficits that remain are specific: near-miss locality on the stress-test matrix (0.639 vs. 0.694) because co-located twins compete inside one retrieval ranking—and that gap closes in the real-capacity setting (Appendix C). Eviction exposure of S8 beliefs is a stress-test phenomenon, not a deployment deficit. Two consequences follow. First, Claim 2’s lifecycle value is *replace and drop*, not editing. Second, the case for internalizing beliefs at this scale rests on design grounds—always-on parametric availability with retrieval disabled, and per-answer attribution—plus stress-test locality, not on the composite. Retrieval-only with a proper lifecycle is the strong default; placement among stores earns its keep at margins the stress-test matrix still measures.

4.3 Persistence stress test

Table 1 is the budgeted-store matrix, read as a persistence stress test. The router scores 0.612 versus oracle 0.639; the gap 0.027 is inside the rerun drift bound 0.025, so we state that the router matches oracle within run-to-run variance, while its margin over the better extreme (0.102) is $4\times$ the drift bound. All-internalize loses reference facts outright—in the failure matrix (Table 2), fact recall in weights spans the low band shown there versus roughly double in the store—and pays on supersession and near-miss locality. All-retrieve fails on evicted items in this stress test, and stale transients stay assertable long after they expire. Random placement inherits both failure families. Dual-write is worst (Table 6): conflict-flagged and plain QA rows both score far below either single-channel reference—the deficit is global, not localized to conflict events; mechanism analysis is out of scope.

4.4 Sequential stability and locality

On a frozen 75-item unrelated pool, the unedited base scores 0.973; all-internalize streams end at 0.893–0.933 after 47–58 sequential edits (Appendix A), consistent with HoReN’s published sequential stability—the codebook backend leaves base weights untouched. Near-miss probes (18 twin pairs) separate cleanly when twins live in different stores; all-internalize suffers most on separability (locality axis, Table 1).

4.5 Failure anatomy

Misrouting is one-directional. All 22 router errors are fact→belief (Table 4): preference-like entries with concrete referents (*favorite X*) are over-internalized—the surface form overlaps the belief criterion boundary in our workload spec. Misrouted items’ QA recall averages 0.272.

Supersession is key competition. On the all-internalize arm, of 23 new-value probes, 12 gate-hit the *old* slot, 7 the new slot (Table 3): consecutive same-subject updates compete in the key space—a general key-value-store phenomenon, not an editing-backend defect—which is exactly why reference facts belong in a store whose records are replaceable.

Trigger is not assertion. Delayed probes of stale transient states fire the corresponding edit at a 0.367-0.400 rate (own-slot 0.367), yet only 0.1 of final answers assert the old state; we report both rates without mechanistic interpretation.

Open-task usage is the weakest link. Scenario probes sit just under the frozen gate: 53/52 of 72 probes land in the [0.85, 0.90) band with fire rates 0.042/0.083 (oracle/router arms). This is a read-path diagnostic—the distance between planner-probe wording and stored keys—not an editing-mechanism deficiency; judged naturalness must be read jointly with usage (e.g., random placement’s high naturalness is hollow fluency at low usage).

5 Related Work

Knowledge editing spans weight-modifying (ROME, MEMIT) and parameter-preserving (GRACE, HoReN) editors [Meng et al., 2022, 2023, Hartvigsen et al., 2023, Fang et al., 2026], with unstructured extensions (UnKE [Deng et al., 2025], AnyEdit [Jiang et al., 2025]) and long-form evaluation (LEME [Rosati et al., 2024]); none decide whether an item should be edited at all, and probes stay question-style. **In-context editing** (IKE [Zheng et al., 2023], MQuAKE/MeLLO [Zhong et al., 2023], DeepEdit [Wang et al., 2024]) decomposes questions against edited memories as text; our elicitation is parametric and our target is open generation. **Memory agents** (MemGPT [Packer et al., 2023], MemoryBank [Zhong et al., 2024], Mem0 [Chhikara et al., 2025]) are exactly the all-external pole we compare against. Closest to us, Memory-R1 [Yan et al., 2025] learns ADD/UPDATE/DELETE/NOOP over a *single external* bank; we decide placement *across heterogeneous stores including weights*, whose irreversibility, interference, and compute cost are what make placement a real decision. Read-side routers (Self-RAG [Asai et al., 2024], Adaptive-RAG [Jeong et al., 2024]) decide *whether to retrieve per query*; we decide *where to write per memory*—orthogonal axes.

6 Limitations and Future Work

Scenario usage rates are low and gate-marginal—a read-path limitation we quantify rather than tune away (no read-path prompt changes after test began). The text-injection ablation is underpowered ($n=19$, null at that scale); parametric elicitation is retained on attribution/persistence design grounds. Results are single-backbone (Llama-3.1-8B-Instruct) on a synthetic self-generated workload of 4 users at ~50 memories each; hidden labels inherit the workload spec’s type boundaries, as the one-directional misrouting shows. The counterfactual utility router’s failure indicates per-memory signals miss lifecycle value; sequence-aware placement learning—training on stream-level replace and drop outcomes—is the indicated next step. Entry-level eviction is a persistence stress test at this scale, not a deployment learning target.

Table 1: Seven-arm main matrix on the full test split (210 memories, 740 probes). Composite = equal-weight mean over recall/freshness/locality per the frozen scoring semantics; CI95 = probe-level bootstrap (1000 draws). Unrelated = 15-item pool (sequential-consistency uses the 75-item pool, App. A). Judged naturalness is always paired with scenario usage (belief/fact keyword rates). [†] preliminary (utility router). [‡] action-space completion arm, appended after the main matrix. Read as a persistence stress test (budgeted oldest-first eviction); the real-capacity reading is Table 5. Drift bound from an identical-config rerun: $|\Delta_{\text{composite}}| = 0.025$.

Arm	Composite	CI95	Recall	Fresh.	Local.	Unrel.	Judge (usage)
ALL-RAG	0.510	[0.434, 0.579]	0.495	0.395	0.639	1.000	0.76 (0.15/0.19)
ALL-EDIT	0.401	[0.319, 0.486]	0.283	0.447	0.472	1.000	0.59 (0.00/0.10)
RANDOM	0.388	[0.304, 0.476]	0.283	0.382	0.500	1.000	0.89 (0.07/0.16)
ORACLE	0.639	[0.564, 0.706]	0.500	0.724	0.694	1.000	0.91 (0.07/0.23)
ROUTER	0.612	[0.548, 0.678]	0.471	0.671	0.694	1.000	0.68 (0.07/0.16)
UTILITY [†]	0.508	[0.432, 0.578]	0.490	0.395	0.639	1.000	0.83 (—)
DUAL-WRITE [‡]	0.226	[0.170, 0.292]	0.119	0.447	0.111	1.000	0.72 (—)

<i>Per-user composite</i>				
Arm	u_3	u_4	u_5	u_6
ALL-RAG	0.598	0.500	0.424	0.518
ALL-EDIT	0.544	0.353	0.360	0.330
RANDOM	0.641	0.301	0.236	0.352
ORACLE	0.788	0.569	0.566	0.624
ROUTER	0.696	0.527	0.593	0.621
UTILITY [†]	0.598	0.495	0.422	0.518
DUAL-WRITE [‡]	0.230	0.280	0.251	0.144

Table 2: Type $\times$ store failure matrix (QA + scenario keyword recall). All arms share one dedup/supersede lifecycle; only the placement decision differs. $\langle \text{arm} \rangle$ subscripts identify the source arm.

Type	RAG	Edit	Drop	Dual
Belief		0.402 ^(S2)		
	0.420 ^(S1)	0.483 ^(S3)		
	0.588 ^(S3)	0.408 ^(S4)	0.062 ^(S3)	0.172 ^(S7)
	0.419 ^(S6)	0.385 ^(S5)		
Fact		0.143 ^(S6)		
	0.527 ^(S1)			
	0.504 ^(S3)	0.233 ^(S2)		
	0.539 ^(S4)	0.268 ^(S3)	0.014 ^(S3)	0.097 ^(S7)
	0.554 ^(S5)	0.282 ^(S5)		
Transient	0.524 ^(S6)			
	0.733 ^(S1)		0.000 ^(S3)	
	0.769 ^(S3)	0.178 ^(S2)	0.000 ^(S4)	0.078 ^(S7)
	0.733 ^(S6)	0.111 ^(S3)	0.000 ^(S5)	

Table 3: Attribution of the ALL-EDIT supersede-new failures (23 probes, gate-only replay): key competition under consecutive same-subject updates — the probe key lands on the old slot although the new key is written and retrievable. Presented as a general key-value-store phenomenon, not an editing-backend defect.

Old slot	New slot	Other row	No hit
12	7	2	2

Table 4: Router confusion against hidden type labels (N=210). All 22 misroutes are one-directional (fact→belief): preference-like entries with concrete referents over-internalized — the favorite-X surface form overlaps the belief criterion boundary (workload spec). Misrouted items’ QA recall averages 0.272.

Hidden	Predicted		
	belief	fact	transient
belief	53	0	0
fact	22	105	0
transient	0	0	30

Table 5: Real-capacity setting (off: budget/eviction disabled; pre-registered P5 item) versus persistence stress test (on: budgeted oldest-first eviction). Distractors remain a read-side noise model when on. Type-aware single-store (S8: beliefs and facts retrieved, transients dropped, zero edits; appended arm). Off is the main reading: all-retrieve recall recovers; S8 is the strongest reported arm, with oracle-level freshness from store-side replacement.

Arm	State	Composite	Recall	Fresh.	Local.
ALL-RAG	on	0.510	0.495	0.395	0.639
ALL-RAG	off	0.624	0.631	0.408	0.833
ROUTER	on	0.612	0.471	0.671	0.694
ROUTER	off	0.708	0.568	0.724	0.833
SINGLE-STORE (S8)	on	0.618	0.490	0.724	0.639
SINGLE-STORE (S8)	off	0.731	0.635	0.724	0.833

Table 6: Dual-write loss decomposition (journal-level). Conflict-flagged and plain rows both score far below either single-channel reference — the deficit is global, not localized to conflict events; mechanism analysis is out of scope. Codebook rows match the analytic expectation per user; edit cost 210 edits / 924.3 s.

Conflict QA	Plain QA	all-edit ref	all-rag ref
0.142 (n=381)	0.080 (n=249)	0.284	0.552

References

- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.
- Jingcheng Deng, Zihao Wei, Liang Pang, Hanxing Ding, Huawei Shen, and Xueqi Cheng. Everything is editable: Extend knowledge editing to unstructured data in large language models. In *International Conference on Learning Representations*, 2025.
- Yuan Fang, Yi Xie, and Xuming Ran. Horen: Normalized hopfield retrieval for large-scale sequential model editing. *arXiv preprint arXiv:2605.08143*, 2026.
- Tom Hartvigsen, Swaminathan Sankaranarayanan, Hamid Palangi, Baekjin Kim, and Marzyeh Ghassemi. Aging with GRACE: Lifelong model editing. In *Advances in Neural Information Processing Systems*, 2023.
- Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong C. Park. Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity. In *Conference of the North American Chapter of the Association for Computational Linguistics*, 2024.
- Houcheng Jiang, Junfeng Fang, Ningyu Zhang, Mingyang Wan, Guojun Ma, Xiang Wang, Xiangnan He, and Tat-Seng Chua. Anyedit: Edit any knowledge encoded in language models. In *International Conference on Machine Learning*, 2025.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. In *International Conference on Learning Representations*, 2023.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.
- Domenic Rosati, Robie Gonzales, Jinkun Chen, Xuemin Yu, Melis Erkan, Yahya Kayani, Satya Deepika Chavatapalli, Frank Rudzicz, and Hassan Sajjad. Long-form evaluation of model editing. In *Conference on Language Modeling*, 2024.
- Yiwei Wang, Muhao Chen, Nanyun Peng, and Kai-Wei Chang. Deepedit: Knowledge editing as decoding with constraints. *arXiv preprint arXiv:2401.10471*, 2024.
- Jing Xu, Arthur Szlam, and Jason Weston. Beyond goldfish memory: Long-term open-domain conversation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, 2022.
- Sikuan Yan, Xiufeng Yang, Zuchao Huang, Ercong Nie, Zifeng Ding, Zonggen Li, Xiaowen Ma, Jinhe Bi, Kristian Kersting, Jeff Z. Pan, Hinrich Schütze, Volker Tresp, and Yunpu Ma. Memory-r1: Enhancing large language model agents to manage and utilize memories via reinforcement learning. *arXiv preprint arXiv:2508.19828*, 2025.
- Saizheng Zhang, Emily Dinan, Jack Urbanek, Arthur Szlam, Douwe Kiela, and Jason Weston. Personalizing dialogue agents: I have a dog, do you have pets too? In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, 2018.
- Ce Zheng, Lei Li, Qingxiu Dong, Yuxuan Fan, Zhiyong Wu, Jingjing Xu, and Baobao Chang. Can we edit factual knowledge by in-context learning? In *Conference on Empirical Methods in Natural Language Processing*, 2023.

Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. Memorybank: Enhancing large language models with long-term memory. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

Zexuan Zhong, Zhengxuan Wu, Christopher D. Manning, Christopher Potts, and Danqi Chen. MQuAKE: Assessing knowledge editing in language models via multi-hop questions. In *Conference on Empirical Methods in Natural Language Processing*, 2023.

A Sequential-Consistency Confirmation

Table 7: 75-item frozen unrelated pool at cumulative-edit checkpoints (gate 0.90). Positioning: reproduction consistent with HoReN’s published sequential stability; no threshold comparison; not merged into composites.

Stream	ck10	ck25	end (edits)
S2-u03	0.973	0.973	0.933 (56)
S2-u04	0.96	0.907	0.907 (49)
S2-u05	0.973	0.973	0.92 (58)
S2-u06	0.92	0.907	0.893 (47)
S5-u03	0.933	—	0.933 (19)
S5-u04	0.96	—	0.947 (18)
S5-u05	0.973	—	0.893 (22)
S5-u06	0.92	—	0.867 (16)
Base spot-check	0.973 (n=75)		

B Gate Threshold Calibration (archive)

Table 8: Dev sweep; the preregistered rule selected 0.90. Calibration archive only — not part of the main narrative.

Threshold	0.75	0.80	0.85	0.90
Own-key hit	1.000	1.000	1.000	1.000
Unrelated false-fire	0.933	0.667	0.267	0.000

C S8 Axis Decomposition

Table 9: S8 versus oracle and router on the frozen five-column scorecard (composite + recall/freshness/locality + unrelated). Off = real-capacity setting; on = persistence stress test. Oracle off was not run.

Arm	State	Comp.	CI95	Recall	Fresh.	Local.	Unrel.
ORACLE	on	0.639	[0.564, 0.706]	0.500	0.724	0.694	1.000
ORACLE	off	—	—	—	—	—	—
ROUTER	on	0.612	[0.548, 0.678]	0.471	0.671	0.694	1.000
ROUTER	off	0.708	[0.654, 0.757]	0.568	0.724	0.833	1.000
SINGLE-STORE	on	0.618	[0.543, 0.686]	0.490	0.724	0.639	1.000
SINGLE-STORE	off	0.731	[0.680, 0.781]	0.635	0.724	0.833	1.000

Table 10: Type $\times$ probe-kind slices requested for S8. Belief cells are empty ($n=0$): supersede and near-miss pairs in test v1.1 fall on facts (workload spec). Fact near-miss is the locality gap (stress-test S8 below oracle/router; gap closes off).

Cell	oracle on	oracle off	router on	router off	single-store on	single-store off
belief $\times$ supersede-old	—	—	—	—	—	—
belief $\times$ supersede-new	—	—	—	—	—	—
belief $\times$ near-miss	—	—	—	—	—	—
fact $\times$ supersede-old	0.304 ($n=23$)	—	0.217 ($n=23$)	0.304 ($n=23$)	0.304 ($n=23$)	0.261 ($n=23$)
fact $\times$ supersede-new	0.783 ($n=23$)	—	0.696 ($n=23$)	0.783 ($n=23$)	0.783 ($n=23$)	0.826 ($n=23$)
fact $\times$ near-miss	0.694 ($n=18$)	—	0.694 ($n=18$)	0.833 ($n=18$)	0.639 ($n=18$)	0.833 ($n=18$)

D Prompt Appendix

Verbatim frozen prompts (generator, router, planner, judge, unrelated-pool) live in the supplementary package under `prompts_verbatim/`, each hash-prefixed. They were frozen before test evaluation and are not reproduced here for space.

E Run Manifests and Freezes

Workload freeze v1.1, the frozen scorecard, the appended S8/analysis freezes, and the S8 axis-decomposition freeze are in the supplementary package. A budget-severity scan was authorized then withdrawn before any run: deployed codebook and retrieval capacity is tens of thousands of records, so entry-level eviction at this workload scale is not a meaningful mechanism. One journal-fragment disclosure: `p3_S7_u03/edits.jsonl` carries one prefix line from an aborted dispatch; the analysis kept the last contiguous segment; `items.jsonl` is unaffected (212 unique keys).